

Professor Franco Bernelli  
 Editor-in-Chief  
 CEAS Space Journal

**Re: Original Research Article** - Observability Limits of Learned Satellite Cyber-Recovery Decisions Under Compromised Evidence

Dear Professor Bernelli and Editors,

I am pleased to submit the manuscript entitled “Observability Limits of Learned Satellite Cyber-Recovery Decisions Under Compromised Evidence” for consideration as an Original Research Article in CEAS Space Journal. I am the sole author and an independent researcher.

The manuscript addresses a space-systems assurance question at the intersection of cybersecurity and artificial intelligence in space. It asks whether a learned satellite cyber-recovery selector can make the correct recovery decision when trusted-looking evidence may itself be false, and how that boundary changes when corroborating evidence is independently informative versus correlated with the compromised path. This fits the journal's coverage of space systems, cybersecurity for space systems, and artificial intelligence in space.

The study reports experiment S7-LSO-001, a separately frozen deterministic finite assurance experiment comprising exactly 1,033 modeled observations. It exhaustively evaluates the complete eight-feature visible-state lattice for a deterministic comparator and visible-only learner, the complete nine-feature lattice for a corroboration-aware learner, and nine prespecified hidden-truth collision observations. The result is intentionally narrow: learning the visible decision boundary does not establish correctness when decisive truth is missing from the learner's inputs. Corroborating evidence can change that boundary, but the unsafe decision returns when the corroborating path shares the same false state. The accepted execution and core outputs are hash-frozen, and a separately implemented repository auditor reconstructs all 1,033 policy decisions with zero mismatches without importing or invoking the primary analyzer.

The manuscript also distinguishes itself from recent CEAS work on spacecraft fault detection, isolation and recovery and AI-supported diagnosis/reconfiguration. Those studies address fault representation, detection, diagnosis, or recovery implementation. Study 7 instead evaluates the trust and information sufficiency of the evidence feeding a cyber-recovery authorization decision under an adversarial evidence model.

For transparency, this manuscript belongs to a broader independent satellite-cybersecurity research program with related manuscripts under consideration elsewhere. None of those manuscripts uses Study 7's 1,033-observation population as experimental evidence. The related-manuscript relationships are summarized below so the editorial office can assess overlap directly.

| <b>Related manuscript</b>           | <b>Evidence boundary</b> | <b>Relationship to this submission</b>                                                                          |
|-------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------|
| JAIS 2026-09-I012066                | Studies 1 and 2          | Response/recovery under contact and adversarial evidence constraints; Study-2 V5 is antecedent motivation only. |
| IEEE TAES                           | Studies 3, 4, and 6      | Temporal evidence, producer composition, and artifact assurance; no Study-7 observations.                       |
| Acta Astronautica AA-D-26-02872     | Study 8                  | Contact-aware cryptographic agility; distinct mechanism and population.                                         |
| Journal of Space Safety Engineering | Separate repository      | Software supply-chain assurance for spacecraft flight software; no Study-7 data, model, or endpoints.           |

The present manuscript is original, has not been published previously, and is not under consideration by another journal. No text, tables, figures, or experimental observations from the related manuscripts are reused as Paper-3 evidence beyond properly cited background or antecedent material.

OpenAI ChatGPT, accessed through the ChatGPT web interface, was used as an interactive assistant during Study 7 development and manuscript preparation. Before the results freeze, it assisted with protocol refinement, code and test drafting, repository and audit documentation, and verification review. After the freeze, it assisted with manuscript organization and drafting, literature-source verification, compliance checking, and language refinement. I made and approved the research decisions, controlled the repository and experimental execution, reviewed the code and text changes, and verified the frozen outputs and numerical claims. This use is disclosed in Methods Section 3.5.

The manuscript includes the required Statements and Declarations, including funding, competing interests, author contributions, ethics/consent, data availability, and code availability. The public repository contains the protocol, implementation, frozen result records, and separately implemented audit materials. The exact accepted Study-7 evidence is permanently archived on Zenodo at <https://doi.org/10.5281/zenodo.22732060> (concept DOI <https://doi.org/10.5281/zenodo.22732059>).

Thank you for considering this work for CEAS Space Journal. I believe its focus on the assurance boundary of learned cyber-recovery decisions will be relevant to readers working on spacecraft autonomy, cybersecurity, fault management, dependable AI, and mission assurance.

Sincerely,

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